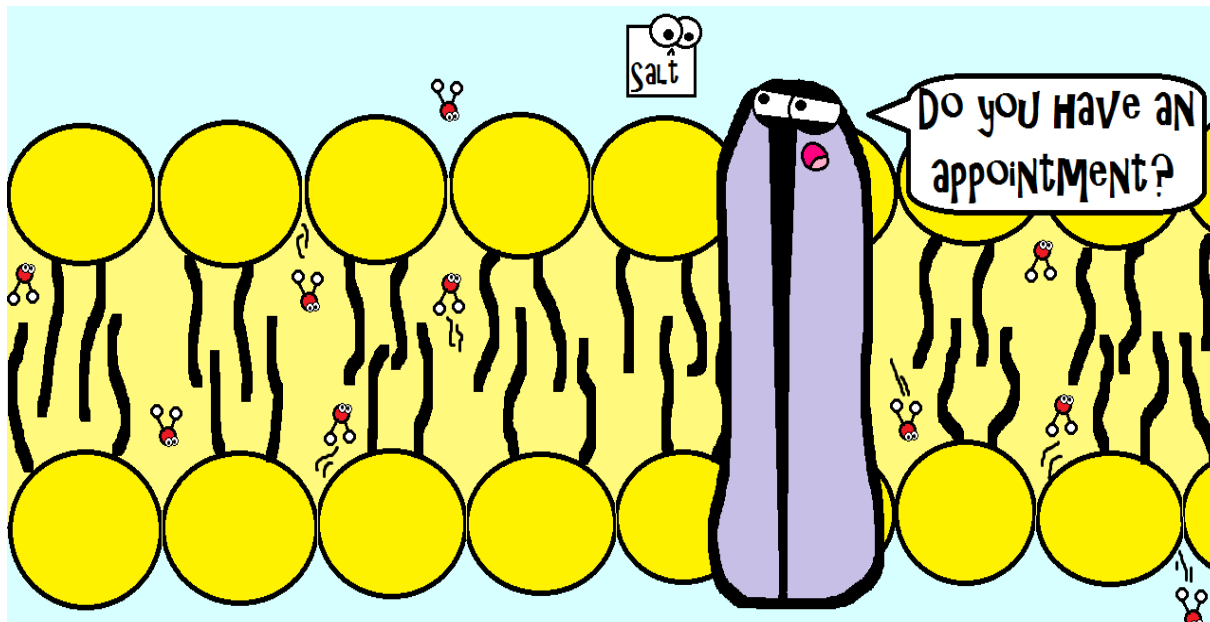


Deciphering the Molecular Basis, Pathophysiology, and Pharmacotherapy of Cystic Fibrosis



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1. Introduction

Cystic fibrosis (CF) is the most prevalent life-limiting autosomal recessive genetic disorder in Caucasian populations. CF is a monogenetic disorder that results from mutations of the Cystic Fibrosis Transmembrane Conductance Regulator (CFTR) gene, which encodes an anion-selective channel. The discovery of the CFTR gene in 1989 started to unveil the underlying molecular basis of CF and provided insights into the pathophysiology of CF which has been crucial for the development of new therapeutic approaches for CF (Kerem et al., 1989). CFTR is located on the apical membrane of epithelial cells of many different tissues and is primarily responsible for the secretion of chloride and bicarbonate (Cl^- and HCO_3^-), for example, into the airways, the lumen of ducts in the pancreas, sweat glands and intestine. The wide distribution of CFTR around the body can lead to multisystem dysfunction when the CFTR is compromised, which also explains the complexity of CF disease (Hanssens et al., 2021). Since the discovery of the CFTR gene till now, there have been 719 CF-causing mutations identified which can be classified based on different defects in CFTR protein biogenesis, folding stability and gating (Zielenski and Tsui, 1995). Among them, the most common mutation is the deletion of phenylalanine 508 (F508del) in nucleotide-binding domain 1 (NBD1) with an occurrence of ~70% of CF alleles in all patients (Cheng et al., 1990). Small molecules called CFTR modulators targeting defective biogenesis and defective gating of CFTR have been developed and are classified into two categories: CFTR correctors which aid folding and trafficking of CFTR protein to the plasma membrane; CFTR potentiators which enhance CFTR channel activity at the plasma membrane hence improving anion secretion. As mutations such as F508del can cause both gating and folding defects hence the combination of correctors and potentiators is necessary for novel pharmacotherapy for CF. This dissertation will discuss the molecular basis of drugs targeting the CFTR protein and investigate its function both as a chloride channel and beyond. The subsequent focus lies on how different mutations impair the normal function of CFTR protein which ultimately leads to multisystem dysfunction. Finally, this dissertation will discuss novel pharmacotherapies against different types of CF based on their genomic profile and elucidate the relationship between personalized medicine and current treatment.

2. Molecular structure and function of CFTR

2.1. CFTR as a member of the ABC family

CFTR is a member of the ATP-binding cassette (ABC) superfamily which is the largest family of transmembrane proteins. Generally, ABC transporters bind to ATP and facilitate the transport of various molecules against a concentration gradient across cell membranes (Dean et al., 2022) whereas CFTR is the only ion channel member of the ABC family and functions uniquely by providing a continuous aqueous pathway for anions to flow passively down their concentration gradient. Typical full ABC transporter functional units consist of two cytoplasmic nucleotide-binding domains (NBDs) responsible for binding and hydrolysis of ATP; two transmembrane domains (TMDs), forming the channel for either importing or exporting substrate. NBDs are characterized by Walker A and Walker B motifs, separated by ~90-120 amino acids, which are the two structurally conserved ATP-binding motifs in all ABC transporters (Hyde et al., 1990). TMDs of ABC transporters usually consist of 6~12 membrane-spanning α -helices which determine substrate specificity (Dean et al., 2022). The human genome contains 48 *ABC* genes of which 44 encode transporter proteins which can be classified into 5 subfamilies (A, B, C, D, and G). So far, mutations in 21 of 48 *ABC* genes are found to be related to diseases including cystic fibrosis (Dean et al., 2022).

CFTR is a 1480 amino acid protein, also termed ABCC7, with a molecular weight of 168.41 kDa (Alam and Locher, 2023). CFTR is composed of a single polypeptide that folds into five domains (TMD1-NBD1-R-TMD2-NBD2 as in Figure 1): two TM domains each consisting of 6 α -helices spanning the membrane which are connected by intracellular and extracellular loops (ICLs & ECLs); two NBDs containing the ATP binding pockets; one regulatory domain (R domain) which contains alternating cluster of positively and negatively charged amino acids, forming consensus sequences for recognition and phosphorylation by cAMP-dependent protein kinase A (PKA) and protein kinase C (PKC) (Riordan et al., 1989).

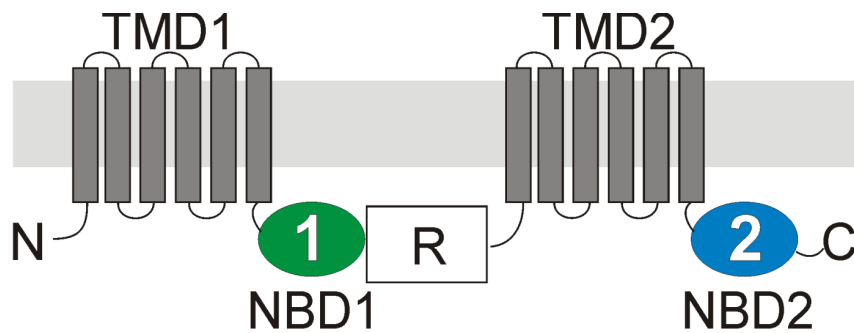


Figure 1 CFTR topology (From Dr. Vergani's slides)

2.2. CFTR mechanism of opening

2.2.1. R domain phosphorylation regulates CFTR channel opening

Phosphorylation of specific consensus amino acids in the R domain stimulates the ATP-dependent channel opening. Evidence has shown that the dephosphorylated R domain has an inhibitory effect on the CFTR opening process as the deletion of the R domain (amino acid 708-835) results in a constitutively active CFTR channel independent of phosphorylation (Rich et al., 1991). Serine 660, 700, 795, and 813 are found to be consensus recognition sites for cAMP-dependent PKA, phosphorylation of these sites upregulates the channel activity (Cheng et al., 1991). What's more, experimental evidence has shown that the phosphorylation signalling is redundant, as mutations of individual serine residues to alanine do not change the responsiveness of CFTR channel activity towards cAMP concentration. Channel activity is impaired only when all four sites are mutated. To conclude, we can say more than one serine site is involved in normal gating whereas none of the serine sites are essential for the activation of the channel (Cheng et al., 1991), and each serine site stimulates the channel activity to a different level. For example, mutations of Ser-660 & Ser-813 decrease the channel open state probability (P_0) to a greater extent compared to other mutations (Ostedgaard et al., 2001). In contrast, it's worth mentioning that the phosphorylation of Ser 737 & 768 is inhibitory hence phosphorylation of these sites downregulates channel activity. Experiments have demonstrated that Ser 768 is the first phosphorylated serine residue and that the P_0 of WT channels is lower than S768A channels in all PKA

concentrations, indicating the damping effect of S768 phosphorylation on channel activity (Csanády et al., 2005).

The exact mechanism of CFTR gating regulation by R-domain phosphorylation is poorly understood. Comparing the cryo-EM structure of the R domain before and after phosphorylation provides a rational explanation of the regulation mechanism (Figure 2). In the dephosphorylated state, the R domain is wedged in between the NBDs and TM helices 9, 10, and 12, inhibiting the dimerization of NBDs as the interface interaction is required for dimerization to occur. In the phosphorylated state, the R domain becomes disordered and is released from its occluded position, moving away from the NBD interface and facilitating dimerization (Zhang et al., 2018, Mihályi et al., 2020).

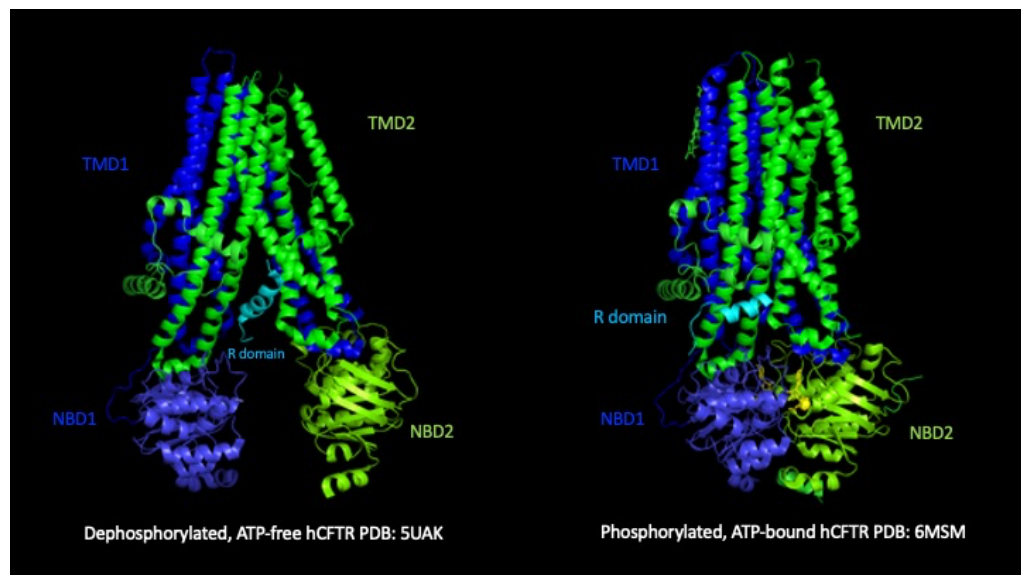


Figure 2 Two states of CFTR protein (generated using PyMOL)

2.2.2. ATP-dependent dimerization of NBDs drives CFTR channel opening

For channel opening, phosphorylation of the R domain is a prerequisite (Ostedgaard et al., 2001) while ATP binding and hydrolysis at the two composite ATP binding pockets are necessary. In dimeric structure, the NBD1 and NBD2 form a rotationally symmetrical ‘head to tail’ conformation, with two ATP molecules at the heterodimer interfaces. Each NBD contains several key motifs essential for ATP binding and hydrolysis: Walker A motif (GXXGXGK-S/T) also termed the P-loop at the “head” region; Walker B motif (R-X₇-hhhhD) at the “head” region; “signature” LSGGQ motif at the

“tail” region. Walker A lysine (K464 in NBD1 & K1250 in NBD2) interacts with the α - or γ -phosphate of the ATP molecule. Mutation of Walker A lysine to alanine inhibits ATP-dependent gating and reduces channel opening time as the closing rate is much faster in mutants than the WT (Powe et al., 2002). Walker B aspartate (D572 & D1370) coordinates the Mg^{2+} in MgATP which is necessary for ATP binding and hydrolysis. The ATP is further stabilized by stacking the π -electron of the adenine ring of the ATP molecule with the side chain of an aromatic amino acid upstream of Walker A (W401 & Y1219) (Hwang and Sheppard, 2009). After docking of the ATP molecules, studies have shown that R555 in the NBD1 tail and T1246 in the NBD2 head interact and form a hydrogen bond which is important for channel opening (Figure 3). Either R555K or T1246N mutations alone slow down the channel opening by disabling the formation of the hydrogen bonds and increasing the energy barrier. However, when the mutations occur simultaneously, the ability to form hydrogen bonds is restored hence the channel open probability is returned to normal (Vergani et al., 2005). The conformational change of the NBD1&2 after ATP binding facilitates the “head to tail” heterodimer formation. After dimerization, the ATP molecule is sandwiched between the Walker A and Walker B motifs of one NBD and the LSGGQ motif of the other NBD. Hydrogen bonds are formed between the oxygen on the γ -phosphate with the side chain hydroxyl groups of S548 and S1546 and also with the nitrogen of G550, G551, and G1349 within the LSGGQ motif (Hwang and Sheppard, 2009). The important role of LSGGQ motif can be further certified by the relatively common disease-causing mutation G551D which abolishes the ATP-dependent gating of the channel. The G551D-CFTR is completely unresponsive towards ATP with a P_o ~100 fold smaller than the WT-CFTR (Bompadre et al., 2008). Experimental data have shown that adding 10 μ M Cd^{2+} in the absence of ATP can increase the activity of G551D-CFTR by 12.2 ± 1.2 -fold ($n = 6$). This indicates that the LSGGQ motif might act as a switch when bound to ligand, either bound with the endogenous ligand ATP or the Cd^{2+} , the rigid body movement of NBDs will facilitate channel opening in TMDs (Wang et al., 2009).

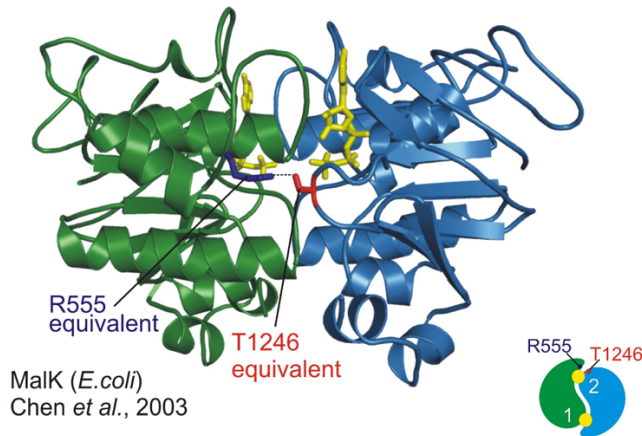


Figure 3 Hydrogen bonds between R555 & T1246 (From Dr. Vergani's slides)

2.2.3. ATP hydrolysis and gating cycles

After dimerization of the NBDs, the channel enters a stable open state (O_1) in which both ATP-binding sites are bound with ATP and the channel allows the flow of ions. The first composite site (NBD1 Walker motifs and NBD2 LSGGQ motif) of the dimer is catalytically inactive, it acts more as a stabilizing site (Aleksandrov *et al.*, 2002). The channel is closed by the hydrolysis of the ATP bound to the second composite site via a short-lived post-hydrolytic O_2 state. Then the dimer dissociates and the channel enters the C_2 state (Csanády *et al.*, 2010). This pathway described is called the hydrolytic closure pathway and it is much faster than the non-hydrolytic closure pathway (O_1 to C_1) in WT-CFTR. Considering the forward and backward reaction rate constants (k_{-1} & k_1 in Figure 4), the channel favors the hydrolytic closure pathway (Langron *et al.*, 2018). For WT-CFTR, the vast majority of the channels (>90%) close via the hydrolytic pathway ($C_1 \rightarrow O_1 \rightarrow O_2 \rightarrow C_2$ in Figure 4). However, for some mutants the channels close via the non-hydrolytic pathway ($C_1 \rightarrow O_1 \rightarrow C_1$ in Figure 4). K1250 and D1370 are Walker A serine and Walker B aspartate which are the two key catalytic residues at composite site 2. In K1250A and D1370N CFTR mutants, prolonged burst duration (τ_b) is observed which is ~140-fold longer than the wild type, suggesting both mutations have completely abolished the hydrolytic activity. This is proved by experiments measuring τ_b on E1371S, K1250A, D1370N and E1371S/K1250A, E1371S/D1370N double mutants. E1371S, K1250A, D1370N single mutations reduce the rate of ATP hydrolysis (k_1) by ~100 fold and increase τ_b while double mutants do

not further reduce k_1 and prolong τ_b . To conclude, both K1250A and D1370N mutants never enter the post-hydrolytic O_2 state and close via the non-hydrolytic pathway (O_1 to C_1) (Langron et al., 2018, Csanády et al., 2010, Gunderson and Kopito, 1995, Simon et al., 2023). K464A mutation at composite site 1 results in a reduced ATP turnover rate of around 10-fold (Ramjeesingh et al., 1999). However, the τ_b for K464A mutants is not significantly affected as the coupling between gating and ATPase cycles is less strict for the mutants compared to WT in which ~80% closes via the non-hydrolytic pathway and ~20% undergoes ATP hydrolysis hence through the O_2 state. This is further attested by experiments on K464A/K1250A double mutant. The double mutant closes ~10-fold faster than the K1250A mutant (Csanády et al., 2010), but more than 1000-fold slower than K464A, suggesting that some K464A mutated channels do undergo the hydrolytic pathway.

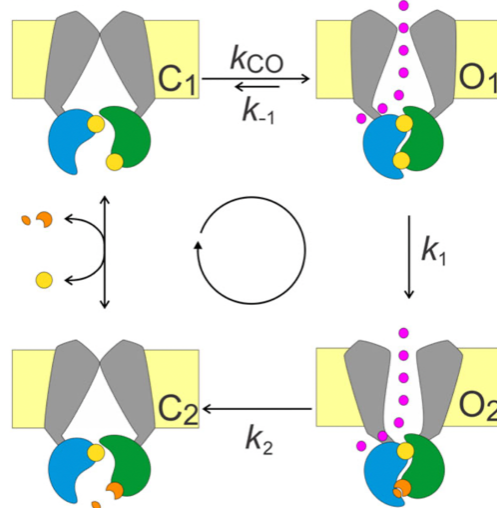


Figure 4 CFTR gating and hydrolytic cycles (Langron et al., 2018)

3. CFTR function and pathophysiology

3.1. CFTR protein's normal function extends its role as a chloride channel and the consequences of dysfunction

3.1.1. Conducting airways

Respiratory system symptoms, including impaired mucociliary clearance (MCC) in which the cilia lining the airway fail to clear the pathogen-containing mucus, chronic lung infections and inflammation that may ultimately lead to lung failure, significantly contribute to the high morbidity

and mortality of individuals with CF. CFTR is expressed at the apical membrane of the epithelial cells lining the conducting airways and in submucosal glands (Hanssens et al., 2021). The apical membrane of the epithelial cells of the airways is lined with a thin layer of fluid called airway surface liquid (ASL). ASL is mainly composed of two layers: the periciliary layer (PCL) which is in direct contact with the epithelial cells and facilitates the beating of the cilia; the mucus layer (MCL) sitting on the top of PCL which contains mucin that can trap inhaled pathogens and remove them from the airway. The normal pH of the ASL is around 7.0~7.6. The maintenance of the ASL composition and relatively neutral pH level is crucial for effective MCC and normal airway function (Zajac et al., 2021). CFTR and epithelial sodium channel (ENaC) function synergistically in this regulatory process. In normal airways, CFTR secretes Cl^- into the lumen in a cAMP-dependent manner and it also causes chloride secretion indirectly by promoting the activity of other chloride channels such as TMEM16A/ANO1 which is a Ca^{2+} -dependent chloride channel. This is achieved by intracellular crosstalk of cAMP and Ca^{2+} signalling pathways (Léris et al., 2018). As the chloride ions enter the lumen of the airway, the osmotic gradient drives concurrent movement of water into the airway leading to ASL hydration. CFTR is not a highly selective anion channel, it also allows the transport of HCO_3^- . When mucin is produced by the goblet cells, it contains Ca^{2+} and H^+ that must be removed for necessary mucin expansion to ~1000-fold and form mucus gel networks (Hanssens et al., 2021). HCO_3^- modulates the rheological property of mucin by reducing Ca^{2+} crosslinks and forming complexes using H^+ (H_2CO_3) hence decreasing the viscosity and increasing the transportability of mucus (Chen et al., 2010). As a result, malfunction of CFTR results in defective chloride and bicarbonate secretion which leads to the production of dehydrated, thick, and viscous mucus that is very difficult to clear. Defective in bicarbonate ion secretion also leads to a decreased pH in CF ASL.

The ENaC is mainly responsible for the reabsorption of Na^+ and it is negatively regulated by CFTR (Stutts et al., 1995). ENaC contains α , β ,

and γ subunit. Its activation requires the cleavage of the extracellular loops of the $\alpha \gamma$ subunit by serine protease/neutrophil elastases (Rossier and Stutts, 2009). To keep the mucus hydrated, the ENaC must be controlled to reduce excessive sodium reuptake. The antimicrobial SPLUNC1 (short palate lung and nasal epithelial clone 1) is a soluble ENaC modulator in the ASL that binds and inhibits ENaC activity. SPLUNC1 binds to ENaC via an 18 amino acid domain and causes ENaC to internalize hence protecting it from being activated by protease cleavage (Zajac et al., 2021). When ASL volume increases, SPLUNC1 is diluted leading to increased absorption of Na^+ and water; when ASL volume decreases, SPLUNC1 is concentrated leading to inhibition of ENaC hence the ASL volume is likely to increase (Hobbs et al., 2012). When CFTR is defective, the ASL becomes acidic (~ 0.5 unit lower than normal) and the acidic environment abolishes the normal function of SPLUNC1. The acidic environment also increases the activity of proteases hence increasing cleavage and activation of ENaC. Both processes result in hyperactivation of the ENaC and disruption of the ASL homeostasis in CF patients (Garland et al., 2013).

3.1.2. Pancreas and Gastrointestinal (GI) tract

CFTR dysfunction also impacts the pancreas and the GI tract. The pancreas is one of the most common organs affected by CF as $\sim 85\%$ of patients suffer from pancreatic insufficiency (PI) while $\sim 15\%$ of patients have remained pancreatic sufficient (PS) (Ahmed et al., 2003). The expression level of CFTR is high in the pancreatic ductal epithelia. In normal conditions, the acinar cells of the pancreas produce highly concentrated digestive enzymes which require dilution along the duct to remain soluble. Impaired CFTR function which causes defective Cl^- and HCO_3^- secretion results in the production of reduced volume and hyper-concentrated pancreatic secretion (Kopelman et al., 1985, Wilschanski and Novak, 2013). The pancreatic juice produced is more acidic than normal: pH decreases from 8.12 ± 0.06 in WT mice to 6.60 ± 0.04 in CFTR^{-/-} mice (Freedman et al., 2001). This abnormal pancreatic secretion can lead to pancreatic obstruction and damage. What's more, CF patients with pancreatic symptoms also encompass complications such as

pancreatitis with the frequency of occurrence depending on the CFTR genotype (Ooi et al., 2011).

With pancreatic duct obstruction, there is insufficient delivery of digestive enzymes into the GI tract. PI-CF patients exhibit compromised digestive function, resulting in symptoms such as steatorrhea and fat malabsorption. They require life-long enzyme supplements to restore normal absorption of nutrients. Duodenum hyperacidity is another hallmark of CF caused by impaired HCO_3^- secretion in which the decreased pH decreases the activity of pancreatic digestive enzymes for both PI and PS patients (Wilschanski and Novak, 2013). Meconium ileus (MI) is an early manifestation of CF within the GI tract. The defective CFTR in the GI tract leads to thick and dehydrated mucus production. HCO_3^- plays a vital role in chelating Ca^{2+} associated with exocytosed, densely packed mucin. This interaction aids the formation of loose, hydrated mucus (Carlyle et al., 2012). When CFTR is defective, the mucin is not disrupted by HCO_3^- resulting in dehydrated and thickened mucus mixed with faecal content that obstruct the ileus, ultimately leading to the occurrence of MI.

3.1.3. Sweat glands

Another hallmark of CF disease is that patients secrete salty sweat which can be used to diagnose CF effectively. CFTR is expressed in both the secretory coil and the absorptive duct of the sweat gland, and uniquely, trafficked to both apical and basolateral membranes (Hanssens et al., 2021). Unlike in any other epithelia, this CFTR expression pattern results in the retention of Cl^- and Na^+ in the lumen leading to excessive production of hypertonic sweat. This diagnostic feature was used by Gibson and Cooke to develop a standardized test for CF in 1959 (Gibson and Cooke, 1959).

3.2. Classifications of CFTR mutations and prevalent pathogenic mutants

Among the more than 2000 identified CFTR mutations, approximately 700 of them have been verified as disease-causing. These mutations have been classified into classes I to VI based on their impact on CFTR structure and

function. Class I mutations result in defective CFTR caused by the translation of unstable, truncated polypeptides due to the introduction of premature termination codons (PTC). Class II mutations lead to defective protein folding and trafficking, with the F508del mutation as an example being the most common CFTR mutation. By assessing CFTR's glycosylation state, studies have shown that F508del CFTR fails to mature to the fully glycosylated state and is degraded at the ER (Cheng et al., 1990). Class III mutations lead to defective gating of CFTR at the membrane with a significantly reduced P_o . Class III includes the G551D mutation located in the LSGGQ motif. Class I-III mutations manifest severe symptoms of CF, attributed to no or little activity of the channel. Class IV mutations contribute to impaired channel conductance. These mutations are often located in the TMDs that assemble the anion permeation pathway of the channel. Class V mutations result in diminished presence of functional CFTR at the membrane due to mis-splicing. Class VI mutations reduce the stability of CFTR at the plasma membrane and CFTR produced have shortened half-lives. Most Class IV-VI mutations retain adequate ability to transport ions hence resulting in mild symptoms, but some exceptions do cause severe CF (Welsh and Smith, 1993, Fanen et al., 2014, Mall et al., 2020).

4. Pharmacotherapy for Cystic Fibrosis

4.1. CFTR potentiators: target defective gating mutations and increase the open probability of CFTR at the cell surface

4.1.1. Ivacaftor

Ivacaftor (VX-770) is the first FDA-approved CFTR potentiator for patients who have G551D or other purely gating mutations in at least one of the alleles. As a potentiator, VX-770 increases the open probability (P_o) of the CFTR channel and increases the CFTR-mediated transepithelial current (Van Goor et al., 2009). VX-770 achieves this by stabilizing the pre-hydrolytic O_1 state. D1370N and K1250R are non-hydrolytic CFTR mutants hence their channel closes via the O_1 to C transition (Figure 4). Experiments have shown that the non-hydrolytic mutant channel P_o is increased by VX-770 to approximately 10-fold. The only possibility for the

potentiation of the mutants is by stabilizing the O₁ state and slowing down the O₁ to C transition. For WT CFTR, VX-770 also potentiates the gating by slowing down the O₁ to O₂ transition. The opening rate for both non-hydrolytic mutant and WT CFTR is not altered (Langron et al., 2018). In both cases, the burst duration of CFTR is prolonged, allowing increased ion flow. For F508del mutants, studies have shown that VX-770 potentiates gating by increasing the ATP-dependent opening rate via stabilization of the transition state. This is achieved by decreasing both hydrolytic and non-hydrolytic closing rates hence resulting in shifting the equilibrium to the open state (Kopeikin et al., 2014). Potentiators are only capable of increasing the ion conductance of activated CFTR channels after they have been correctly folded and activated by PKA. Consequently, potentiators alone can only rescue defective gating CFTR mutations and require combination with the correctors to rescue defective folding mutations such as F508del.

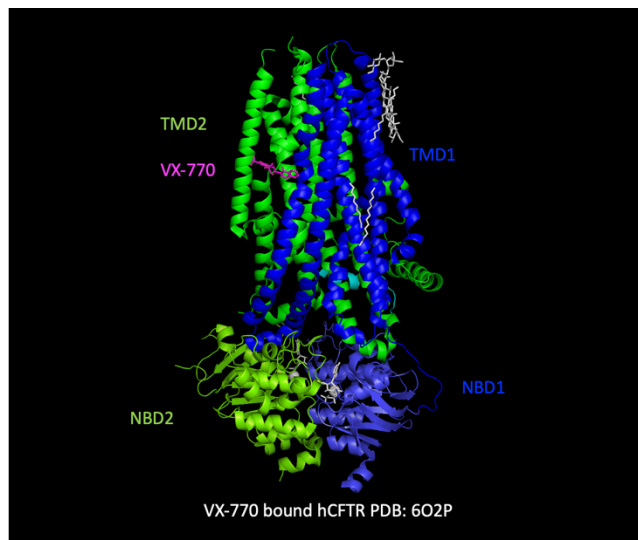


Figure 5 VX-770 bound CFTR (generated using PyMOL)

4.2. CFTR correctors: target misfolding and trafficking mutations and increase the abundance of functional CFTR at the cell surface

4.2.1. Essential role of Phe-508 in CFTR folding and channel gating

Phe-508 deletion is observed in at least one allele of up to 90% of CF patients. Structural investigations have shown that Phe-508 contributes largely to the molecular contacts between NBD1 and ICL4 of TMD2. After assembly and folding of the NBD1, NBD1 docks to the integrated TMD2

after TMD2 translation, this interaction stabilizes both TMD2 and NBD1. Based on the 3D structure of CFTR (Figure 6), NBD1 and ICL1&4 of TMD2 are in close proximity. To pinpoint the exact contact site of the NBD1/CL4 interface, cysteine cross-linking is used. Experimental findings have revealed that the Phe-508 can cross-link with multiple cysteine sites introduced in ICL4 (Leu-1065/Phe-1068/Gly-1079/Phe-1074). Notably, this cross-link can only be found in mature CFTR protein which indicates the importance of NBD1/ICL4 interaction for CFTR maturation (Serohijos et al., 2008). Studies also have shown that F508del disrupts the post-translational folding of NBD2, thereby preventing NBD2 from attaining its trypsin resistant conformation. This assumption is validated by using increasing concentrations of trypsin to detect the proteolytic susceptibility of NBD2, revealing an approximately 100-fold increase in proteolytic susceptibility for NBD2 in the presence of the F508del mutation (Du et al., 2005). Misfolded F508del CFTR lacks complex glycosylation at 37 °C resulting in it being trapped in the ER, ubiquitinated and finally degraded by the proteosomes (Prins et al., 2022).

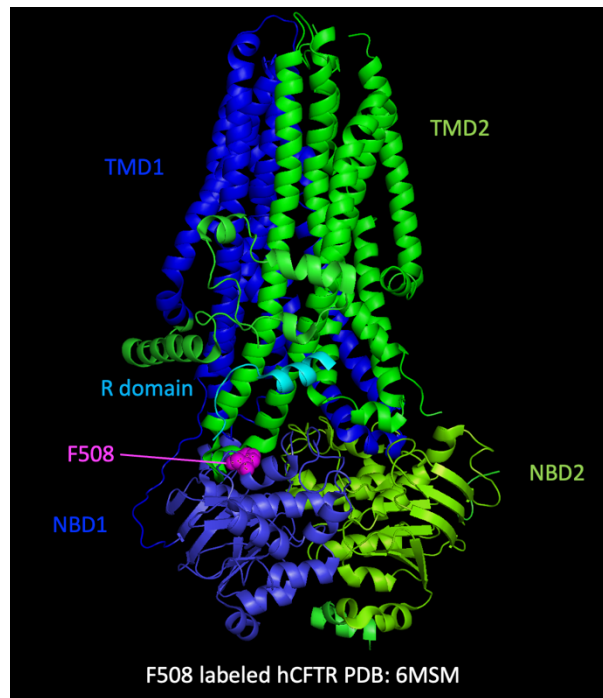


Figure 6 F508 in CFTR (generated using PyMOL)

4.2.2. Lumacaftor and Tezacaftor (Type I correctors)

Lumacaftor (VX-809) and Tezacaftor (VX-661) are classified as type I correctors and are thought to share an overlapping binding site, as when

they are in combination, there is no additive rescue effect. This has also been attested by the competition binding assay: the binding of [^3H]lumacaftor is measured with an increased concentration of Tezacaftor and the results demonstrate a 1:1 competitive mechanism (Fiedorczuk and Chen, 2022a). Type I correctors share structural similarities, characterized by the presence of BCC (1,3-Benzodioxol-5-yl-Cyclopropane Carboxamide) head group. This group is essential for the accommodation of the corrector into its binding site in TMD1. The binding site is formed by TM1, 2, 3, and 6 and the BCC head interacts with S364, L195, and A198 by van der Waals interaction (Figure 7)(Fiedorczuk and Chen, 2022a).

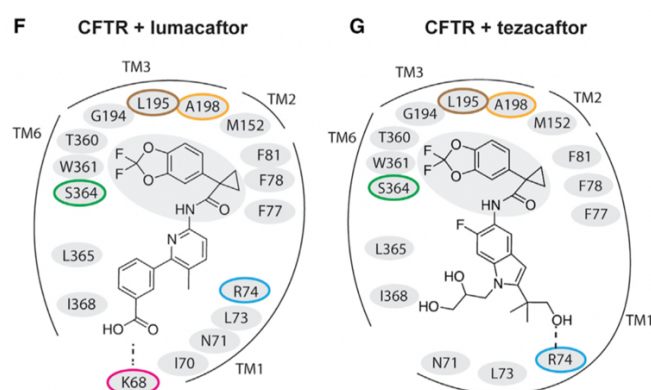


Figure 7 Lumacaftor and Tezacaftor binding pocket (Fiedorczuk and Chen, 2022a)

Lumacaftor and Tezacaftor correct the folding and assembly defects of CFTR by stabilizing TMD1. CFTR 380, truncated after 380 amino acids, thus only containing the TMD1, is the minimal length required for VX-809 action. Although residues 371-375 are not in direct contact with the VX-809, they interact with the N-terminal of TM1 at residue N66 and P67 and facilitate the formation of the binding site. Experiments have shown that F374A/L375A double mutants severely impact the ability of VX-809 to rescue defective CFTR (Ren et al., 2013). Through the application of algorithms predicting the thermodynamic stability of membrane insertion, we can determine that TM1, 2, 3, and 6 are not stable at the early stage of folding as their Gibbs free energy for membrane insertion is positive. Lumacaftor, as well as Tezacaftor, stabilizes TMD1 by filling the cavity and structurally linking the TMs hence lowering the Gibbs free energy to a negative value, facilitating the folding and insertion process (Fiedorczuk

and Chen, 2022a). As a result, Lumacaftor is very effective in rescuing folding defects arising from mutations within TMD1. Given the highly cooperative nature of the folding of CFTR, by stabilizing TMD1, lumacaftor can also allosterically rescue other mutations, including F508del. However, it is not surprising that its effectiveness is somewhat limited for F508del-CFTR. The enhanced stability of the CFTR results in decreased proteolytic sensitivity and less probability of being recognized as defective by the ER and Plasma Membrane quality control mechanism (Serohijos et al., 2008). Nevertheless, it only restores the channel activity of homozygous F508del CF patients to ~15% of normal function hence combination therapy with other correctors or potentiators is necessary (Van Goor et al., 2011).

4.2.3. Elexacaftor (Type III corrector)

Elexacaftor (VX-445) is classified as a type III corrector that binds to the interface of TM 2, 10, 11 and the lasso motif and allosterically stabilizes NBD1 (Fiedorczuk and Chen, 2022b, Veit et al., 2020). Besides being a corrector, VX-445 also has an acute potentiation effect independent of co-incubation time which increases P_0 . The ICL4 between TM helices 10 and 11 of TMD2 interact with NBD1 and form the NBD1/TMD2 interface which is crucial for both NBD assembly and the propagation of signals from NBDs to TMDs to allow channel opening. One key residue in this interaction is R1102 located on TM11 where the Elexacaftor forms a hydrogen bond and a salt bridge. R1102A mutation largely impedes both the correction and potentiation effect of Elexacaftor. A study using gel-shift assay (only mature CFTR is complex-glycosylated, while immature CFTR located in the ER only has a small amount of core-glycosylation) shows that Elexacaftor's correction ability is significantly diminished when R1102A is present. Furthermore, by measuring the macroscopic current across the membrane, the potentiation of Elexacaftor on F508del-CFTR is found to be significant but R1102A-CFTR shows no response towards Elexacaftor (Fiedorczuk and Chen, 2022b). The fact that Elexacaftor is a corrector and potentiator simultaneously makes it a perfect component for combination therapy with other modulators.

4.3. Synergy effect of combination therapy composed of folding correctors and gating potentiators provides novel CFTR treatment

4.3.1. Trikafta

Trikafta, the combination of two folding correctors VX-661 & VX-445 and the gating potentiator VX-770, is approved by FDA in 2019 (and EMA and MHRA shortly after) for CF patients carrying F508del on at least one allele. VX-661 and VX-445 synergistically rescue F508del-CFTR. This has been demonstrated by assessing the PM density of F508del-CFTR with different drugs. VX-661 and VX-445 alone increase F508del-CFTR PM density to 3.5% and 15% of the WT respectively. When in combination, the F508del-CFTR PM density is increased to 45% of the WT, with an EC₅₀ of 0.28 μ M (Veit et al., 2020). VX-445 and VX-661 are also capable of rescuing rare defective folding mutants. Experiments are conducted on 10 CFTR mutants that lead to at least 95% reduction of PM density. Among these, 8 of them (S13F, 13C, E92K, R352G, V520F, L1077P) show a significant increase in correction efficacy compared to single agent correctors, the PM density reaches the therapeutically significant levels of >20% of WT (Veit et al., 2020). Clinical data has also confirmed the efficacy of Trikafta in F508del patients. In a 24-week, phase 3 clinical trial involving 403 patients who are 12 years and older with at least 1 F508del allele: 14.3% higher in FEV₁ indicates an improved lung function; the rate of pulmonary exacerbation is decreased by 63%; chloride concentration in sweat is 41.8 mmol lower than the control group (Middleton et al., 2019). However, evidence has shown that the third component VX-770 diminishes the correction effect of type II CFTR mutants by VX-661 and VX-445. This is demonstrated by performing western blot and immunofluorescence on CFBE-F508del cells with or without ivacaftor. Although the amount of band for mature F508del CFTR is increased with both triple-modulator and double-modulator treatment, the amount of band for mature F508del CFTR is significantly reduced when VX-770 is added compared to VX-661 and VX-445 only (Becq et al., 2022). What's more, the destabilization of F508del CFTR by VX-770 is in a dose-dependent manner. For WT or

G551D, long-term exposure to VX-770 does not affect the CFTR PM density (Veit et al., 2014).

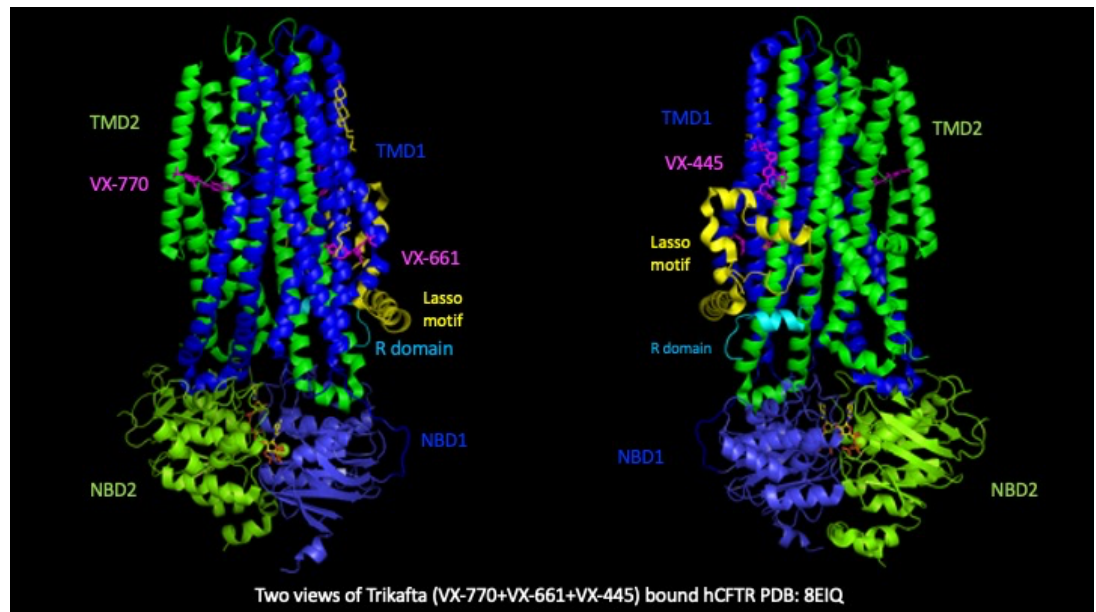


Figure 8 Trikafta bound CFTR (generated using PyMOL)

4.4. Precision medicine in CFTR treatment

Precision medicine, also known as personalized medicine, is a novel approach to CF treatment that considers an individual's characteristics including genotype, environment, and lifestyle. Early diagnosis through newborn screening is crucial for CF. Once diagnosed, genetic analysis of individual patients identifies specific CFTR mutations that guide the use of specific drugs to restore the normal function of CFTR. For example, Trikafta is used for F508del mutations and VX-770 is used alone for G551D mutations. Due to individual differences, even patients with the same genotype may demonstrate different symptoms and that's when personalized treatment strategies are necessary. For example, lung infections can be caused by different microorganisms and require personalized antibiotics; enzyme supplements are needed by PI patients whereas PS patients do not; for patients with CF-related diabetes mellitus, early diagnosis and intervention can significantly reduce mortality (Kelly and Moran, 2013). By interpreting individuals' responses to specific drugs, treatment plans can be modified with the progress of the disease, and ultimately results in the improvement of patient-specific treatment (Paranjape and Mogayzel, 2018).

5. Discussion and Conclusion

In recent years, advances in understanding the structure of CFTR have led to a better understanding of the pathophysiology of CF and shed light on the mechanism underlying CFTR folding and trafficking. Although the molecular details are well-studied, there are still gaps that remain in the field, for example, the exact mechanism of how VX-770 decreases the correction effect of the correctors remains unclear and the mechanisms of action behind different drugs require further investigation. Addressing these gaps could help to refine the CFTR treatments.

With the development of novel drugs, CF is gradually transforming into a chronic disease in adulthood. The extended life expectancy opens a new personalized treatment approach for rare CFTR mutations. By utilizing cultured cells (biopsies) from patients, we can regenerate tissue organoids in vitro using iPSC (induced pluripotent stem cells). These organoids possess the same genomic profile as the patient, offering a platform for drug screening before administering them to the patient (Sachs et al., 2019). For rare mutations resulting from early PTC leading to complete absence of CFTR biogenesis, mRNA therapies are the future of the research area.

Currently, CF patients face significant pharmacological burdens as multiple medications are required to target multisystem symptoms. This situation is believed to be improved by the introduction of modulators as they alleviate the symptoms and potentially reduce the number of medications needed. With the development of innovative treatment approaches, I believe the transformation of CF to a manageable chronic disease is within reach.

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